

DUSHYANT MAHAJAN

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Education

Northeastern University

Master of Science, Computer Software Engineering | **GPA:** 3.67/4.00

Sep 2022 - Dec 2024

Boston, MA

Skills

Core ML Skills: NLP, RAG, Agents, Transformers, BERT, GPT, Diffusion models, Reinforcement Learning, Computer vision, Deep Learning, MLOps, Statistics, OpenCV, Segmentation, Multimodal, GAN

Cloud & MLOps: AWS, GCP, Vertex AI, Docker, Kubernetes, Airflow, GitHub

Programming and Tools: Python, Go, PyTorch, AutoGen, Scikit-Learn, SQL, MySQL, PostgreSQL, MongoDB, OpenCV, Pandas, NumPy

Professional Experience

OptiMe Health, Boston, MA

Apr 2025 – Present

AI Engineer [Amazon Bedrock, RDS, EC2, Redis, DynamoDB, SNS, Lambda, IAM, GitHub Actions]

- Led a team of Software Engineers to develop and deploy **agentic recommendation** pipeline parsing symptoms and emotions, improving user session duration 25% and reducing LLM insight generation time 40%
- Built 30+ RESTful APIs for LLM integration on AWS, leveraging **Flask** and **Redis** for scalable, event-driven architecture
- Deployed serverless AWS Lambda microservices for **AI-driven document enhancement**, optimizing user-specific workflows
- Integrated secure backend endpoints in Node.js to handle Stripe based subscription workflows via **webhooks**, integrating **rate limiting** for AI services and enforcing access controls to ensure compliant billing and usage
- Developed an educational chatbot using **RAG**, boosting user engagement metrics (completion rates) by 30% through personalized guidance
- Automated cross-region AWS service deployments with **GitHub Actions**, cutting deployment time by 20+ minutes

Raga AI, Fremont, CA

Jan 2024 - Jul 2024

Data Scientist [HuggingFace, Langchain, LlamaIndex, Crew AI, Vertex AI, Docker, DSPy, Pydantic]

- Engineered LLM evaluation framework with 50+ **metrics/guardrails**, boosting LLM response quality and accuracy for production
- Collaborated to **open-source** Raga AI Catalyst, an **observability** platform that aids **RAG and agentic systems** debugging, facilitating to identify and resolve performance bottlenecks, earning **15k+ GitHub stars**
- Architected RAG Builder to unify document loaders, LLM APIs, and vector databases, speeding up testing and rapid prototyping
- Tested RAG/agentic workflows across OpenAI, Vertex AI, Ollama, and Groq, optimizing prompts and benchmarks
- Created **Docker**-based CI/CD pipelines with **ML Flow** and **DVC** for robust ML model testing and deployment

Askim Technologies, Mumbai, India

May 2020 – May 2022

Founding Software Engineer [ReactJS, NodeJS, MongoDB, AWS, Python, Golang, Packer, Pulumi, SQL]

- Led **DevOps** initiatives by architecting and deploying a highly available MERN stack application on **AWS**, implementing multi-AZ infrastructure that achieved 99.9% uptime
- Automated infrastructure provisioning using **Pulumi** (IaC) for **AWS** services (VPC, EC2, RDS, Lambda), ensuring consistent, scalable, and reliable cloud environments.
- Integrated Docker containers with AWS **Lambda** functions for **serverless** processing of image and video generation pipeline
- Increased click-through rate by 15% through **A/B testing** UI/UX features utilizing **CloudWatch** logging metrics for monitoring
- Streamlined an automated AMI generation pipeline using HashiCorp Packer and **GitHub Actions**, establishing a reliable CI/CD process that reduced deployment errors and infrastructure updates

Projects

Reducing EHR (Electronic Health Record) Chart Burden using LLMs | [ACL Paper Link](#)

- Researched and improved a LLM pipeline for summarizing **Electronic Health Records** (EHRs) from MIMIC-IV dataset, reducing hallucinations, and improving chunk retrieval accuracy utilizing prompt engineering and context-length optimization
- Engineered RAG workflows with robust QA framework, attaining **86% F1 BERT Score** using locally hosted **Llama models**, reducing clinician chart review times by 30% and maintaining **patient data privacy**

Optimized Fine-Tuning of LLMs on Apple Silicon (MLX + LoRA) | [Link](#)

- Fine-tuned the Ministral-8B-Instruct model on a medical symptoms and diagnosis dataset, reducing test **perplexity** from 2.01 to 1.42 and boosting response accuracy by 60% versus the baseline model
- Implemented LoRA and QLoRA (4-bit quantization) on MLX, achieving 65% lower memory usage during training, and **2x faster inference** speeds on Apple silicon